Trend Following with EMA Crossovers and an ADX Filter

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CFA Club IIT Indore – Induction Task 2026, Finance, Track 2

Abstract

This report evaluates a long-only moving-average trend-following strategy on Bajaj Finance, Titan, and ITC using daily adjusted prices from 2015 to 2025. Parameters were selected using the development period, 2015–2021, and evaluated on a held-out 2022–2025 test period. The selected EMA 10/50, ADX > 15 strategy produced mixed development results and negative cumulative returns for two of three stocks in the test period. It underperformed buy-and-hold in every development stock and in two of three test stocks.

1 Strategy idea and stock selection

The strategy seeks sustained price trends using a fast and a slow exponential moving average (EMA). An ADX filter is added to the filtered version to avoid entering when measured trend strength is low. Bajaj Finance, Titan, and ITC were chosen to represent three liquid, well-known Indian equities with different price behaviour. Selecting stocks known to have performed well over 2015–2025 introduces hindsight and survivorship-style selection risk; results on these stocks do not establish that the method would have worked on a universe chosen in advance.

2 Data

Daily OHLCV data for Bajaj Finance, Titan, ITC, and the Nifty 50 index were loaded from Yahoo Finance using `yfinance`. The notebook requests 2015-01-01 through 2026-01-01, with the end date exclusive; the observed series cover 2015–2025. Adjusted prices are used to account for distributions and corporate actions reflected in the downloaded series. Daily data suit a medium-term crossover rule while permitting next-session execution and daily stop checks.

The notebook checks missing values, duplicate dates, zero-volume days, non-positive prices, OHLC consistency, gaps longer than five calendar days, and daily moves above 10%. It removes duplicate dates, sorts dates, drops rows missing OHLC values, removes rows with non-positive OHLC prices, and fills missing volume with zero. The actual before/after check counts are not saved in the supplied result files: [**TODO: report cleaning-check counts and explain material gaps or zero-volume days from notebook output**].

Corporate-action details supplied in the brief are Bajaj Finance's 2016 split and bonus and 2025 bonus, ITC's 1:2 bonus in 2016 and ITC Hotels demerger in January 2025, and no split or bonus for Titan during the period. These are not independently sourced in the supplied files: [**TODO: add source for each corporate-action statement**].

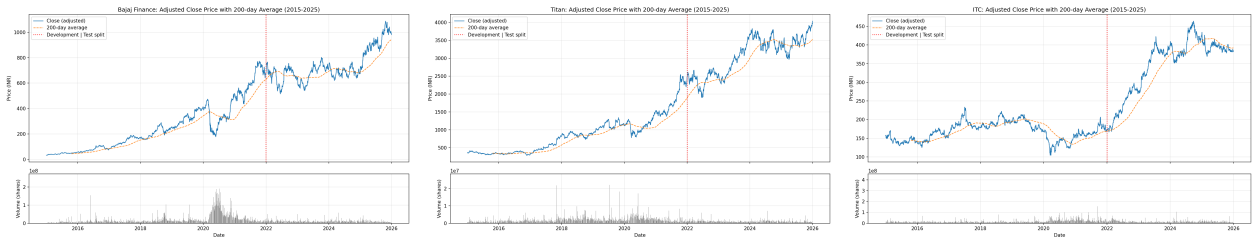


Figure 1: Adjusted closing prices, 200-day simple moving averages, and development/test split. The split falls at the end of 2021.

3 Indicators and trading rules

Let C_t, H_t, L_t, O_t denote adjusted close, high, low, and open on day t. The final parameters are EMA fast = 10, EMA slow = 50, ATR period = 14, ADX period = 14, and ADX threshold = 15. The notebook's exponential averages use `adjust=False` and require a full-period warm-up.

1. **EMA:** $EMA_t^{(n)} = \alpha C_t + (1-\alpha) EMA_{t-1}^{(n)}$, where $\alpha = 2/(n+1)$.
2. **ATR:** $TR_t = \max(H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|)$. ATR is the Wilder-smoothed true range over 14 periods.
3. **ADX:** Directional movement is computed from changes in highs and lows; $+DI$ and $-DI$ are Wilder-smoothed directional movement divided by Wilder-smoothed true range, times 100. $DX = 100|+DI-(-DI)|/(+DI+(-DI))$, and ADX is Wilder-smoothed DX, with period 14. Wilder smoothing uses $\alpha = 1/14$.
4. **Entry signal:** Basic is long when the fast EMA exceeds the slow EMA. ADX-filtered is long when the fast EMA exceeds the slow EMA and ADX exceeds 15. The target is shifted by one session, so a close signal determines the next session's position.
5. **Exit signal:** Exit when the applicable long condition turns false. Signals are executed at the next session's open.
6. **Position and risk:** Long-only; buy the maximum whole shares affordable with available cash, including entry cost. The trailing stop is the highest close reached less $3 \times ATR$, where the stop uses the prior session's ATR and is checked against today's low. A stop-out blocks re-entry until the target signal turns off, then resets.
7. **Execution costs:** Entry price is open increased by slippage; exit price is open (or stop execution price) reduced by slippage. Fees are charged on each side.

4 Backtest assumptions

Each stock is tested independently with INR 1,000,000 initial capital, whole shares, 0.10% fee and 0.05% slippage per side. Open positions are closed at each period's end; each test period starts flat. Buy-and-hold buys at the first open and sells at the final close using the same costs. Nifty 50 is rebased to the same initial capital and dates without costs because the index itself is not directly purchased in this implementation.

The notebook assumes a 6% annual risk-free rate and uses daily equity returns. Annualised return is geometric over elapsed calendar years. Sharpe uses daily excess return over the daily rate equivalent to 6%, divided by sample standard deviation, multiplied by $\sqrt{252}$. Sortino uses the mean excess return divided by the root mean square of negative excess returns, also multiplied by $\sqrt{252}$. Maximum drawdown is the minimum percentage decline from a prior equity peak. Undefined trade statistics are shown as N/A; profit factor is undefined when there are no losing trades, while average win/loss is undefined when there are no trades of that type.

The notebook code contains causality checks comparing truncated and full-history indicators at multiple cutoffs, signal-shift assertions, trade-period and trade-order assertions, equity/PL reconciliation, open-price execution checks, flat-day checks, and a zero-cost-versus-cost check. Final-run sanity assertions are present in the notebook, but saved execution output confirming they passed was not supplied: [**TODO: confirm from the final notebook run that all look-ahead and sanity checks passed**].

5 Parameter selection and split

The development grid covers fast EMA $\{10, 20, 30\}$, slow EMA $\{50, 100, 150\}$, and ADX threshold $\{15, 20, 25, 30, 35\}$; a Basic no-ADX result is also computed. The score is mean Sharpe across the three stocks in development. The notebook selects the setting with the highest neighbourhood-smoothed score, averaging nearby grid settings across fast, slow, and ADX dimensions to favour a plateau over an isolated high value. The final selection is EMA 10/50 and ADX > 15. The supplied plot shows the

raw development mean Sharpe grid, with selected cell outlined.

Development is 2015-01-02 to 2021-12-31; test is 2022-01-03 to 2025-12-31. The test period is run after parameter selection in the notebook's later cells. The notebook code indicates development-only selection; saved execution history confirming the test was run only once was not included.

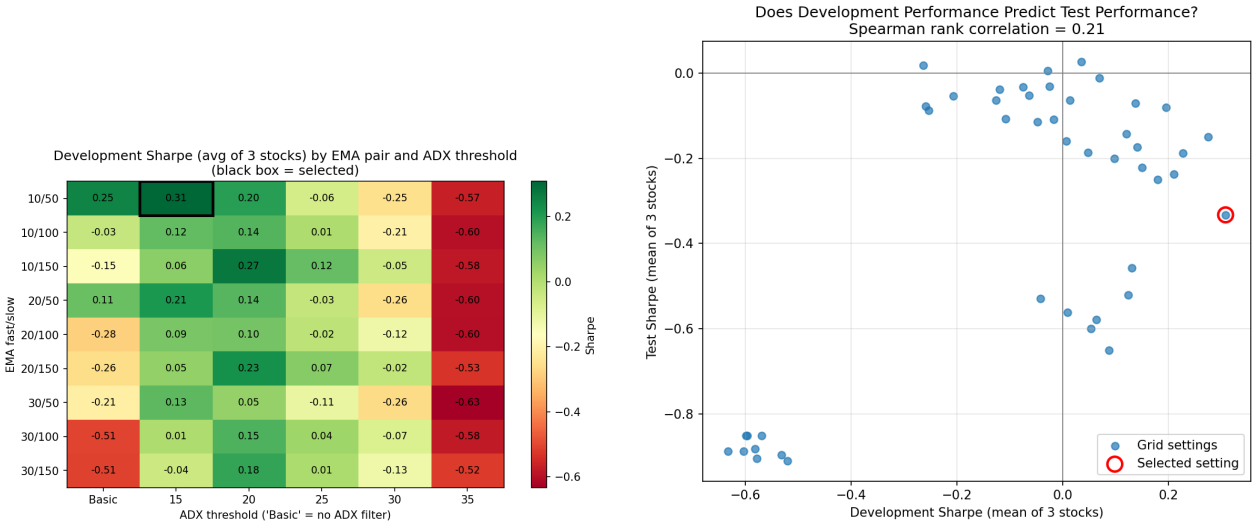


Figure 2: Development Sharpe grid (left) and development-versus-test performance across settings (right). Rank correlation is only 0.21, indicating weak rank persistence.

6 Results

Tables 1 and 2 report the supplied performance metrics. Returns and trade outcomes are after costs. The benchmark return is listed alongside each strategy row for comparison; excess is strategy cumulative return less the Nifty 50 return. Buy-and-hold uses the same costs as the strategy. The tables include cumulative and annualised return, Sharpe, Sortino, maximum drawdown (MDD), trade count, win rate, profit factor (PF), mean winning and losing P&L, and best and worst trade P&L. Values are rounded to two decimals as requested.

Table 1: Development results, 2015-01-02 to 2021-12-31. The ADX-filtered strategy beat Nifty for Bajaj Finance only; buy-and-hold beat both strategies for Titan and did not materially grow ITC.

Stock	Method	Cum.%	Ann.%	Sh.	So.	MDD%	Trades	Win%	PF	Avg+ INR
Bajaj	Basic	249.75	19.59	.79	1.43	-15.88	12	66.67	12.96	338280.73
	ADX	402.68	25.95	.95	1.64	-19.15	26	57.69	7.33	310850.77
	Hold	1927.39	53.73	1.17	1.73	-62.44	N/A	N/A	N/A	N/A
Titan	Basic	85.31	9.22	.26	.46	-23.95	17	58.82	2.40	146124.30
	ADX	67.22	7.62	.18	.31	-25.56	18	55.56	2.08	129501.69
	Hold	584.17	31.63	.84	1.31	-41.73	N/A	N/A	N/A	N/A
ITC	Basic	9.29	1.28	-.30	-.44	-20.14	20	30.00	1.23	81810.99
	ADX	14.23	1.92	-.21	-.30	-24.84	27	40.74	1.36	49232.18
	Hold	9.04	1.24	-.03	-.04	-55.28	N/A	N/A	N/A	N/A
Nifty 50 (all stocks)	No costs	109.37	11.14	.37	.50	-38.44	N/A	N/A	N/A	N/A

Nifty return is 109.37% for every row; strategy-minus-Nifty (percentage points) is, by stock and method: Bajaj Basic 140.38, ADX 293.31, Hold 1818.02; Titan Basic -24.06, ADX -42.15, Hold 474.80; ITC Basic -100.08, ADX -95.14, Hold -100.33. N/A means no applicable trade statistics or undefined statistic. Buy-and-hold and index do not have completed strategy trades.

Table 2: Test results, 2022-01-03 to 2025-12-31. Both EMA strategies lost money in Bajaj Finance and Titan and substantially lagged the benchmark; ITC ADX-filtered was positive but lagged buy-and-hold and Nifty.

Stock	Method	Cum.%	Ann.%	Sh.	So.	MDD%	Trades	Win%	PF	Avg+ INR
Bajaj	Basic	-1.23	-.31	-.34	-.53	-31.05	15	33.33	.97	93441.10
	ADX	-1.44	-.36	-.36	-.57	-27.36	20	30.00	.97	85864.98
	Hold	43.76	9.51	.26	.38	-32.71	N/A	N/A	N/A	N/A
Titan	Basic	-1.94	-.49	-.42	-.59	-25.81	18	33.33	.95	66812.45
	ADX	-1.68	-.42	-.47	-.67	-23.10	18	38.89	.96	58433.66
	Hold	62.99	13.01	.40	.59	-28.60	N/A	N/A	N/A	N/A
ITC	Basic	-10.49	-2.74	-.73	-1.00	-22.71	12	16.67	.68	112730.15
	ADX	13.44	3.21	-.16	-.23	-14.45	15	46.67	1.45	61849.33
	Hold	123.12	22.25	.89	1.37	-20.32	N/A	N/A	N/A	N/A
Nifty 50 (all stocks)	No costs	50.28	10.74	.41	.57	-16.47	N/A	N/A	N/A	N/A

Nifty return is 50.28% for every row; strategy-minus-Nifty (percentage points) is, by stock and method: Bajaj Basic -51.52, ADX -51.72, Hold -6.52; Titan Basic -52.22, ADX -51.96, Hold 12.71; ITC Basic -60.77, ADX -36.84, Hold 72.84. N/A means no applicable trade statistics or undefined statistic. Buy-and-hold and index do not have completed strategy trades.

Development performance is strongest for Bajaj Finance, where both strategies exceed the index but remain below buy-and-hold. The ADX-filtered approach is weak for Titan and ITC relative to the index. In the test period, the trend strategies fail to capture buy-and-hold gains in Bajaj Finance and Titan; only ITC's filtered strategy has a positive result, yet it trails both ITC buy-and-hold and Nifty. These test results do not support a broad claim of profitability or benchmark outperformance.

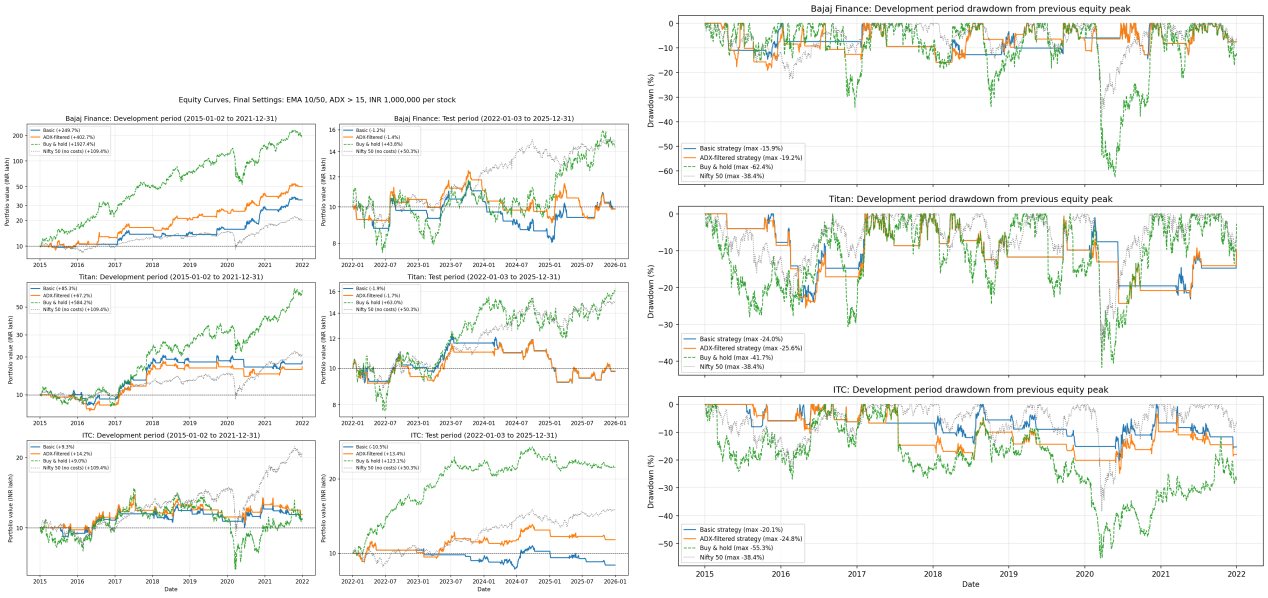


Figure 3: Final-setting equity curves and development drawdowns. Buy-and-hold grew much more for several stocks, while the trend strategies had smaller drawdowns in some cases.

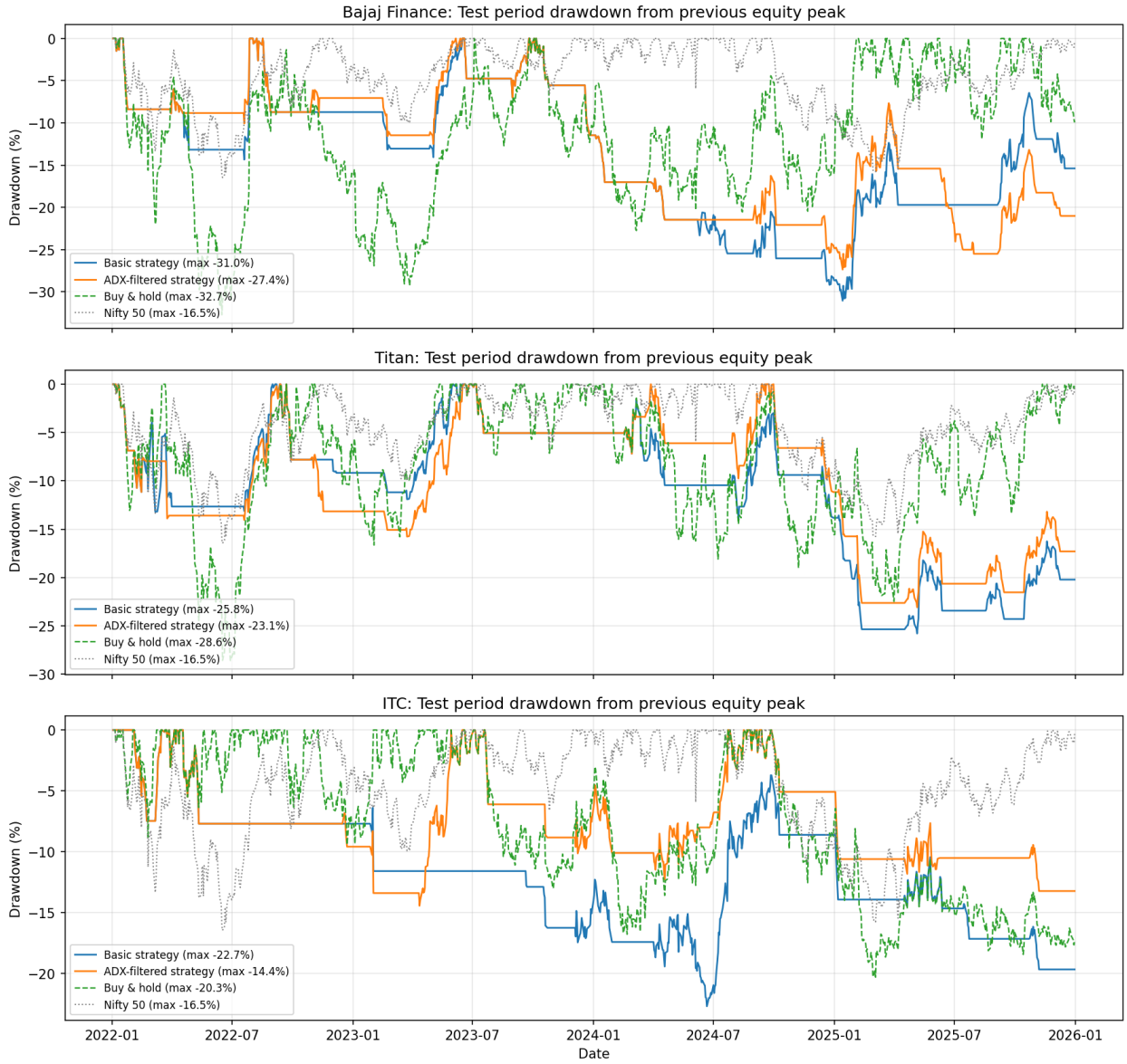


Figure 4: Test-period drawdowns. The Nifty 50 drawdown was smaller than the strategies' drawdown for Bajaj Finance and Titan; ITC ADX-filtered had the smallest drawdown of its shown comparisons.

7 Additional analysis

7.1 Market regimes

Regimes are identified using the prior close: a 200-day simple moving average and ADX threshold 20 classify rising, sideways, or falling conditions. The table gives pooled-period example results for the selected strategy, using cumulative return within each stock-period-regime segment.

Table 3: ADX-filtered cumulative returns by selected regimes. Sideways markets were difficult in test for all three stocks.

Period	Stock	Rising %	Sideways %	Falling %
Development	Bajaj Finance	139.77	71.18	24.71
Development	Titan	54.58	7.13	3.83
Development	ITC	8.07	9.40	-4.63
Test	Bajaj Finance	-3.38	-10.11	13.48
Test	Titan	19.22	-16.87	-0.79
Test	ITC	24.00	-3.42	-5.28

The regime results vary materially by stock and period: test-period sideways returns were negative for all three. Regime-conditioned returns are descriptive, not evidence that the regime rule itself was selected or validated as a trading signal.

7.2 Whipsaws and stop-outs

Table 4: Losing streaks and stop statistics for the ADX-filtered strategy. Stop exits were common, particularly during development.

Period	Stock	Trades	Longest losing streak	Stopped out %	$\leq$ 10-day trades %
Development	Bajaj Finance	26	2	61.54	34.62
Development	Titan	18	4	72.22	22.22
Development	ITC	27	5	44.44	40.74
Test	Bajaj Finance	20	5	50.00	30.00
Test	Titan	18	3	55.56	33.33
Test	ITC	15	2	53.33	46.67

Repeated entries and exits in sideways conditions are consistent with crossover whipsaws, although short duration alone does not prove a trade was a whipsaw. The supplied table reports losing streaks and stop exits, but does not provide counts of short trades; this strategy is long-only. [**TODO: if the task specifically requires a short-trade analysis, explain that short trades are not part of this system.**]

7.3 Profit concentration

The top three trades accounted for 55.52% of gross profit for development Bajaj Finance ADX-filtered and 54.85% for development ITC ADX-filtered; removing the top three would leave reported compounded returns of 146.12% and −11.80%, respectively. For development Titan ADX-filtered, removing its top three changes the return to −20.58%, from 67.22%. In the test period, returns without the top three are negative for Bajaj Finance and Titan Basic/ADX and for ITC ADX-filtered (−21.21%); results therefore depend on a small number of favourable trades in some samples.

7.4 Signal lag and distinction from noise

The rule distinguishes a trend from noise only through EMA ordering and ADX strength; it cannot identify a trend in real time without lag. Median test entries occurred 32.50–48.00 calendar days after the measured prior low, and the price move already gone at entry was 61.67%–85.42% for Basic and 62.99%–75.85% for ADX-filtered. Median captured move was negative for all six test groups, and the strategy gave back 33.04%–59.85% of the measured move by exit. These diagnostics show material lag and do not establish an optimal noise filter.

7.5 Stability and costs

Selected-setting mean Sharpe across stocks declined from 0.31 in development to −0.33 in test; Spearman rank correlation across settings was 0.21. The selected ADX setting had test Sharpes of −0.36 (Bajaj Finance), −0.47 (Titan), and −0.16 (ITC). Neighbouring settings had mixed test results, so parameter stability is weak. Raising combined cost and slippage from 0.15% to 0.30% per side pair reduced selected-strategy returns by 78.51 percentage points for development Bajaj Finance, 18.09 for development Titan, 18.53 for development ITC, and 11.84, 10.60, and 10.21 points for the three test stocks, respectively. Costs materially affect results.

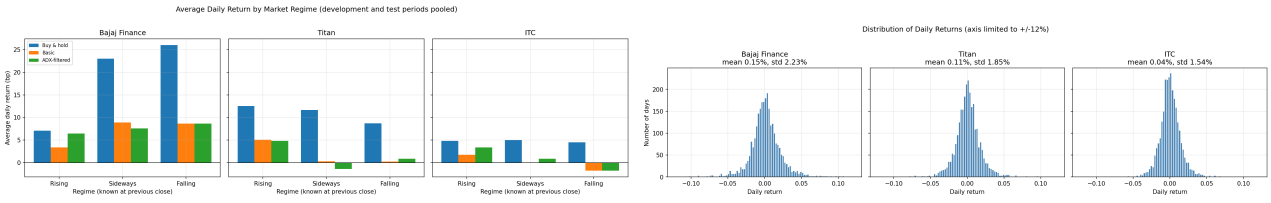


Figure 5: Average daily strategy returns by prior-close regime (pooled periods) and daily return distributions. Both show that average outcomes differ by stock and market state.

8 Limitations and possible improvements

- The three stocks were selected with knowledge of their 2015–2025 histories; test results do not remove this selection bias.
- The sample contains only three equities, and the strategy is long-only.
- Whole-share sizing leaves residual cash and may not generalise to different capital levels.
- Costs and slippage are assumptions; the sensitivity table shows meaningful impact.
- The selected setting performs poorly out of sample and development performance weakly predicts test ranking.
- Yahoo Finance data can be revised, and corporate-action handling should be verified against sourced records.
- The notebook's development defaults differ from its selected final parameters; reports should distinguish exploratory defaults from the frozen setting.
- Improvements could include a broader pre-defined stock universe, walk-forward evaluation, alternative noise filters, and more realistic execution modelling. These are proposals, not tested results.

9 Resources and acknowledgements

Data source: Yahoo Finance accessed through `yfinance`. Analysis uses `pandas`, `numpy`, and `matplotlib`. ATR and ADX are attributed to J. Welles Wilder Jr., *New Concepts in Technical Trading Systems* (1978). [**TODO: add any tutorials or articles used, or state that none were used**]

AI tools Claude and ChatGPT helped with coding and drafting. I can explain the code and results presented here.

A Trade charts

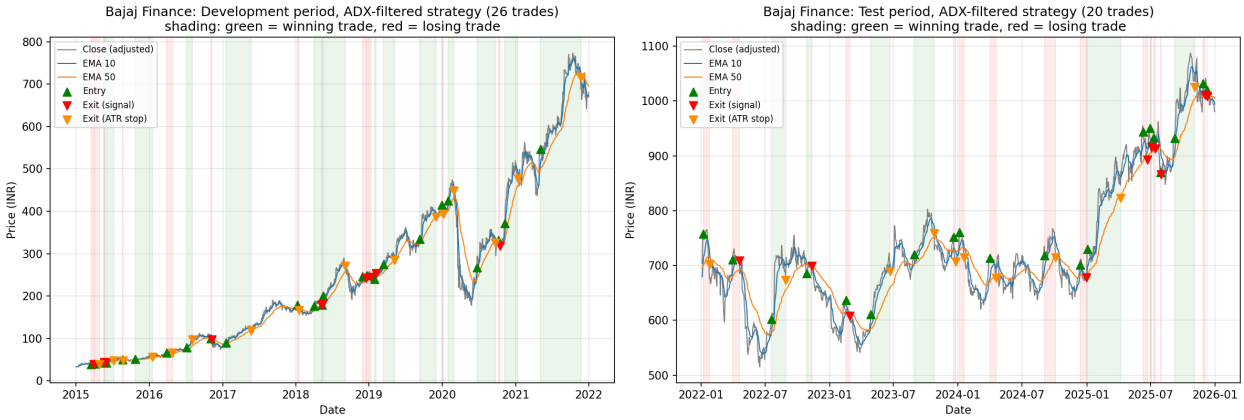


Figure 6: Bajaj Finance trade markers for the ADX-filtered strategy in development and test. Shading distinguishes winning and losing trades.

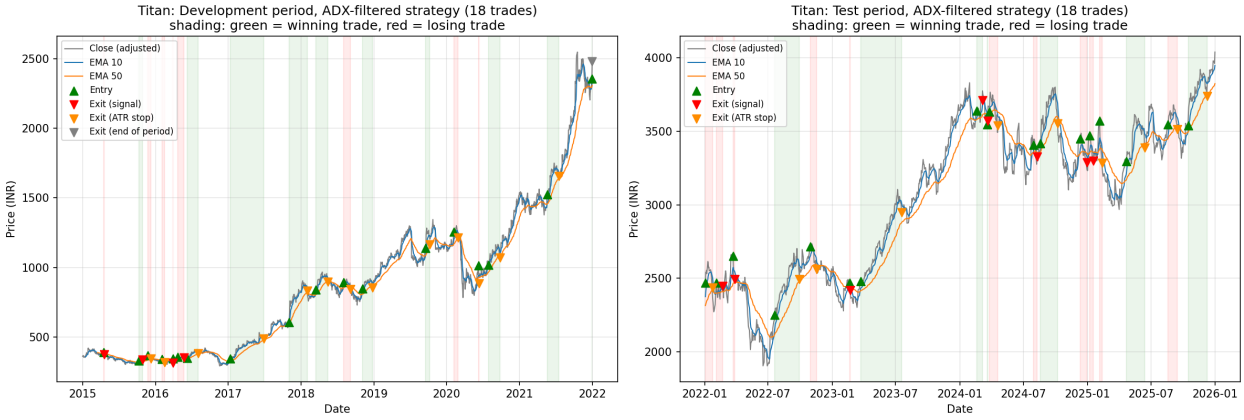


Figure 7: Titan trade markers for the ADX-filtered strategy. The charts show frequent reversals and stop exits as well as sustained trend periods.

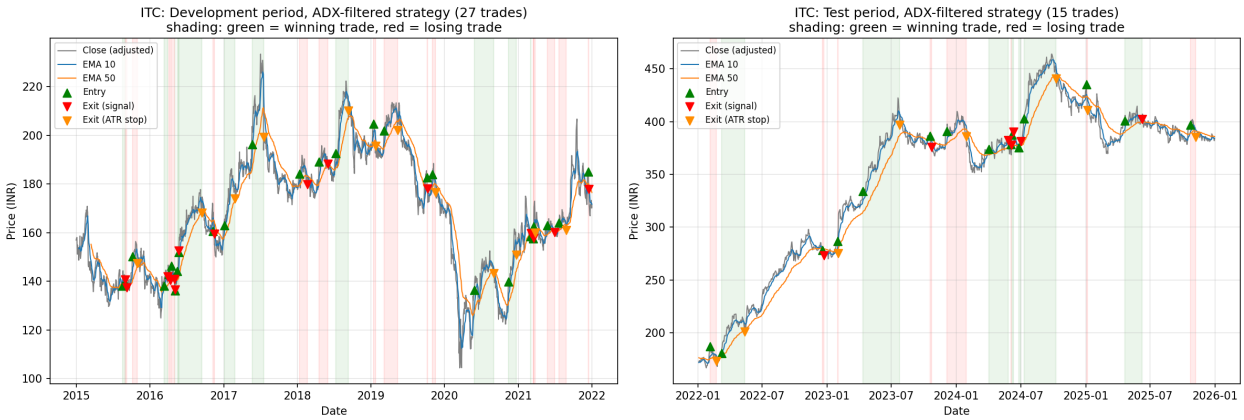


Figure 8: ITC trade markers for the ADX-filtered strategy. Test-period gains were positive for the filtered method despite negative daily-return risk ratios.